

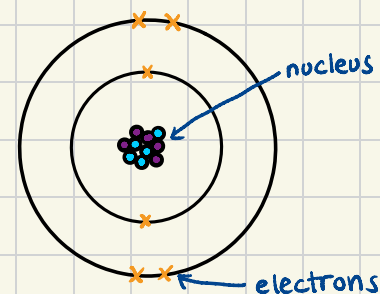
1 - Atomic structure

03/12/24

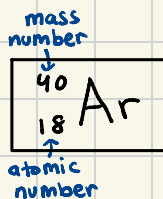
→ atoms are made from 3 subatomic particles

	Location	mass	charge
Protons	inner shell	1	+1
Neutrons	inner shell	1	0
Electrons	outer shell	$\frac{1}{1840}$	-1

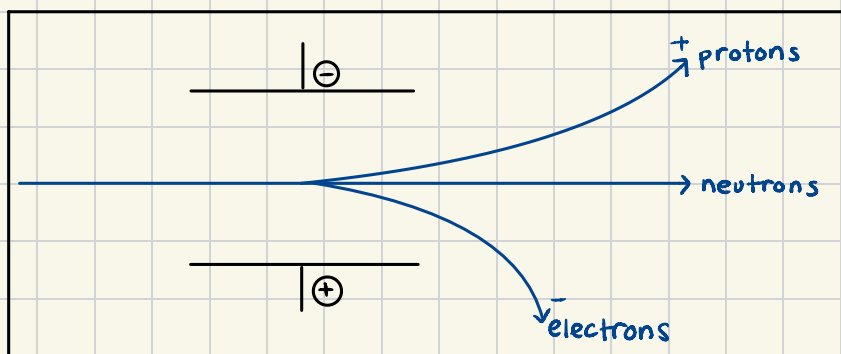
* an atom is mainly empty space



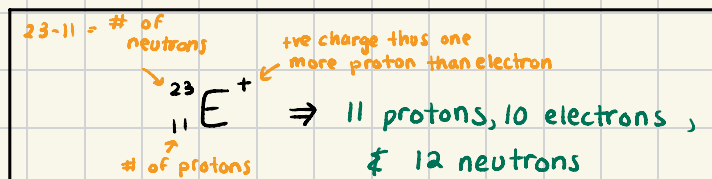
⇒ the nucleus is extremely small but it's where nearly all of the mass of the atom is



- atomic number is the number of protons in the nucleus of an atom
- mass number is the sum of the number of protons & neutrons in the nucleus



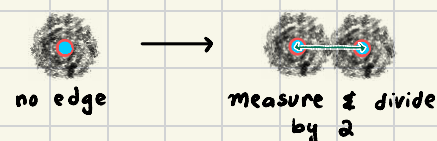
⇒ when put in an electric field, electrons deflect more than protons because its mass is extremely light



Atomic and Ionic radius

⇒ there is no definitive 'edge' to an atom

⇒ the atomic/ionic radius is half the distance between two adjacent atoms/ions



atomic radius

! atomic radius decreases across the period
due to gradual increase in positive nuclear charge which increases the attraction of valence electrons to nucleus

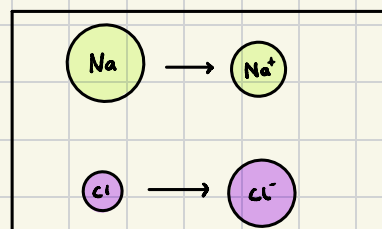
! atomic radius increases down the group
extra shells are being added & the inner shells shield valence electrons which decreases their attraction to the nucleus

	Li	across the period →						
	Na	Mg	Al	Si	P	S	Cl	Ar
↓ down the group	K							
	Rb							

ionic radius

! ionic radius decreases as cationic charge increases
as the net positive charge increases, there are the same number of protons attracting less electrons which increases attraction between electrons & nucleus

! ionic radius increases as anionic charge increases
the increase in the number of electrons w/o increase in nuclear charge weakens the attraction between the valence electrons & nucleus



Isotopes

- * atoms with the same atomic number (protons) but with different mass numbers
 - * isotopes of the same element have the same chemical properties but different physical properties (mass & density)
- eg. ${}^{12}_6\text{C} \neq {}^{14}_6\text{C}$, ${}^{24}_{12}\text{Mg} \neq {}^{26}_{12}\text{Mg}$

Electronic structure

- * electrons move around the nucleus in regions of the atoms called SHELLS or ENERGY LEVELS
- * each shell is given a 'principal quantum number'
- * the further a shell is from the nucleus, the higher its energy and principal quantum number

max number of electrons in each energy level = $2n^2$

↳ first shell ($n=1$): $2(1)^2 = 2$ electrons

↳ second shell ($n=2$): $2(2)^2 = 8$ electrons

↳ third shell ($n=3$): $2(3)^2 = 18$ electrons

- shells are divided into subshells

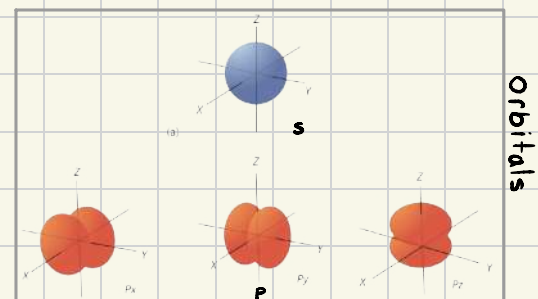
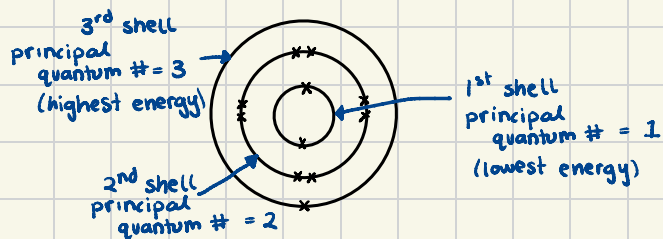
↳ there are 4 kinds of subshells: s, p, d, f ← in terms of energy [s < p < d < f]

- subshells contain one or more orbitals

↳ each orbital can be occupied by a maximum of 2 electrons

★ an orbital is just a region with very high probability of finding electrons

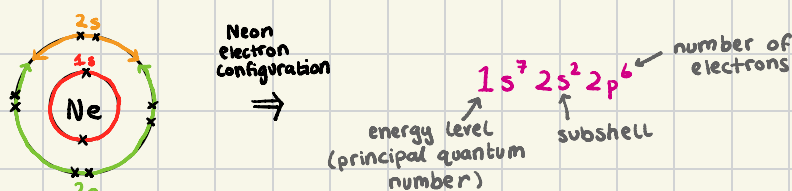
s	: 1 orbital	(1 × 2 = max 2 electrons)
p	: 3 orbitals	(3 × 2 = max 6 electrons)
d	: 5 orbitals	(5 × 2 = max 10 electrons)
f	: 7 orbitals	(7 × 2 = max 14 electrons)



- the ground state is the most stable electronic configuration of atom which has the lowest amount of energy
- this is achieved by filling the subshells with the lowest energy first (1s)

Electronic configuration

* the number of electrons that an atom has & how they are arranged



1 H Hydrogen 1.008	2 He Helium 4.003																																									
3 Li Lithium 6.941	4 Be Beryllium 9.012											5 B Boron 10.811	6 C Carbon 12.011	7 N Nitrogen 14.007	8 O Oxygen 15.999	9 F Fluorine 18.998	10 Ne Neon 20.180																									
11 Na Sodium 22.990	12 Mg Magnesium 24.305											13 Al Aluminum 26.982	14 Si Silicon 28.086	15 P Phosphorus 30.974	16 S Sulfur 32.066	17 Cl Chlorine 35.453	18 Ar Argon 39.948																									
19 K Potassium 39.098	20 Ca Calcium 40.078	21 Sc Scandium 44.956	22 Ti Titanium 47.867	23 V Vanadium 50.942	24 Cr Chromium 51.996	25 Mn Manganese 54.938	26 Fe Iron 55.845	27 Co Cobalt 58.933	28 Ni Nickel 58.693	29 Cu Copper 63.546	30 Zn Zinc 65.38	31 Ga Gallium 69.723	32 Ge Germanium 72.631	33 As Arsenic 74.922	34 Se Selenium 78.971	35 Br Bromine 79.904	36 Kr Krypton 84.798																									
37 Rb Rubidium 84.468	38 Sr Strontium 87.62	39 Y Yttrium 88.906	40 Zr Zirconium 91.224	41 Nb Niobium 92.906	42 Mo Molybdenum 95.95	43 Tc Technetium 98.907	44 Ru Ruthenium 101.07	45 Rh Rhodium 102.906	46 Pd Palladium 106.42	47 Ag Silver 107.868	48 Cd Cadmium 112.414	49 In Indium 114.818	50 Sn Tin 118.711	51 Sb Antimony 121.760	52 Te Tellurium 127.6	53 I Iodine 126.904	54 Xe Xenon 131.29																									
55 Cs Cesium 132.905	56 Ba Barium 137.328	57-71 Lanthanides	72 Hf Hafnium 178.49	73 Ta Tantalum 180.948	74 W Tungsten 183.84	75 Re Rhenium 186.207	76 Os Osmium 190.23	77 Ir Iridium 192.227	78 Pt Platinum 195.085	79 Au Gold 196.967	80 Hg Mercury 200.592	81 Tl Thallium 204.383	82 Pb Lead 207.2	83 Bi Bismuth 208.980	84 Po Polonium [209]	85 At Astatine [210]	86 Rn Radon [222]																									
87 Fr Francium [223]	88 Ra Radium [226]	89-103 Actinides	104 Rf Rutherfordium [261]	105 Db Dubnium [262]	106 Sg Seaborgium [266]	107 Bh Bohrium [264]	108 Hs Hassium [269]	109 Mt Meitnerium [278]	110 Ds Darmstadtium [281]	111 Rg Roentgenium [280]	112 Cn Copernicium [285]	113 Nh Nihonium [286]	114 Fl Flerovium [289]	115 Mc Moscovium [289]	116 Lv Livermorium [293]	117 Ts Tennessine [294]	118 Og Oganesson [294]																									
																		57 La Lanthanum 138.905	58 Ce Cerium 140.116	59 Pr Praseodymium 140.908	60 Nd Neodymium 144.242	61 Pm Promethium [144.913]	62 Sm Samarium 150.36	63 Eu Europium 151.964	64 Gd Gadolinium 157.25	65 Tb Terbium 158.925	66 Dy Dysprosium 162.500	67 Ho Holmium 164.930	68 Er Erbium 167.259	69 Tm Thulium 168.934	70 Yb Ytterbium 173.055	71 Lu Lutetium 174.967										
																		89 Ac Actinium [227.028]	90 Th Thorium 232.038	91 Pa Protactinium 231.036	92 U Uranium 238.029	93 Np Neptunium 237.048	94 Pu Plutonium 244.064	95 Am Americium 243.061	96 Cm Curium 247.070	97 Bk Berkelium 247.070	98 Cf Californium 251.080	99 Es Einsteinium [254]	100 Fm Fermium 257.095	101 Md Mendelevium 258.1	102 No Nobelium 259.101	103 Lr Lawrencium [262]										

! noble gases can be used as checkpoints in writing the electronic configuration

Rules

① aufbau principle

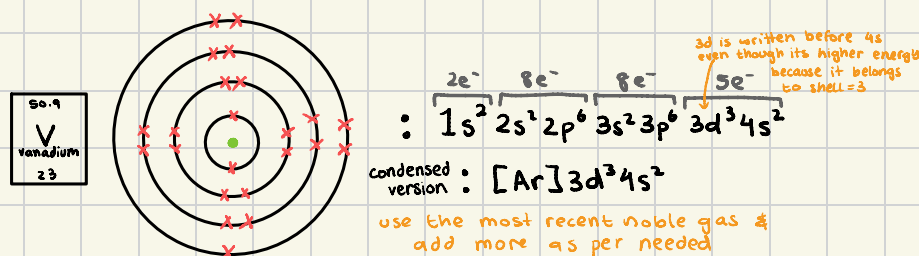
electrons always fill up the lowest energy orbitals first ($s < p < d$)

② Hund's rule

electrons don't pair up in the same orbitals until all other orbitals have been occupied

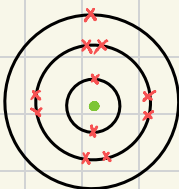
③ Pauli exclusion principle

electrons that are paired in the same orbitals have opposite spins (↑↓) due to repulsion

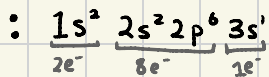


eg.

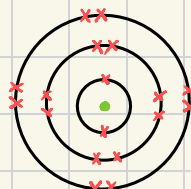
23.0
Na
Sodium
11



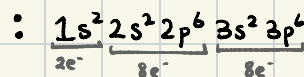
condensed: [Ne] 3s¹



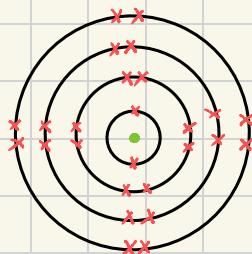
39.9
Ar
Argon
18



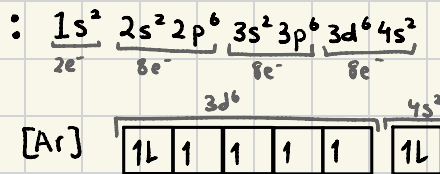
condensed: [Ar]



55.8
Fe
Iron
26

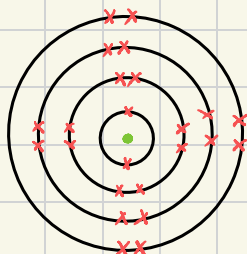


condensed: [Ar] 3d⁶ 4s²



⇒ chromium & copper

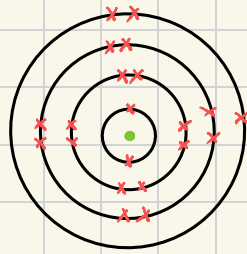
52
Cr
Chromium
24



expected: [Ar] 3d⁴ 4s²

actual: [Ar] 3d⁵ 4s¹

63.5
Cu
Copper
29



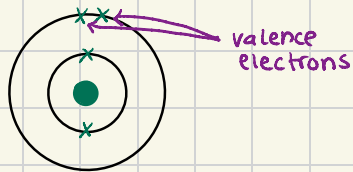
expected: [Ar] 3d⁹ 4s²

actual: [Ar] 3d¹⁰ 4s¹

* chromium & copper have special electronic configurations because achieving half-filled (3d⁵) or fully filled (3d¹⁰) d-subshell provides **extra stability** due to reduced electron-electron repulsion & increased symmetry

Free radical

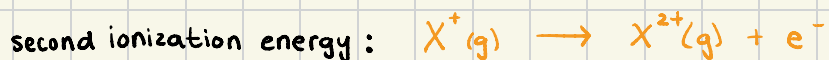
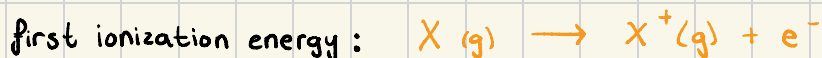
* an atom, molecule, or ion that has atleast one unpaired valence/free electron



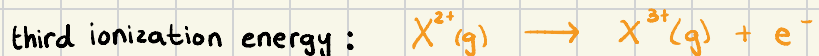
First ionization energy (IE)

06/12/24

! the energy required to remove one mole of electrons from one mole of gaseous atoms to form one mole of gaseous +1 ions



← energy required to remove one mole of electrons from one mole of gaseous +1 ions → +2 ions



← energy required to remove one mole of electrons from one mole of gaseous +2 ions → +3 ions

* IE is required to break the attraction between the nucleus & outermost electron

→ Factors that affect ionisation energy

① atomic radius

→ further away an electron is from the nucleus, easier it is to remove it ; lower IE

② Shielding

→ more shells between the outermost electron & nucleus, easier it is to remove ; lower IE

③ nuclear charge

→ higher the nuclear charge (more protons), more attraction & reduced atomic radius ; higher IE

④ spin-pair repulsion

→ if an orbital is full (2 electrons), they repel each other thus easier to remove the electron ; lower IE

→ Ionization energy trends

• across a period

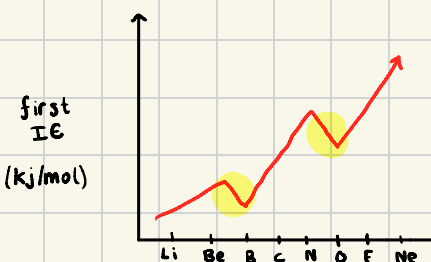
→ atomic radius ↓ [smaller atomic radius]

→ shielding = ^{no} effect same shielding

→ charge ↑ more protons

} increases IE

TREND : IE increases down the group



Be → B

Boron requires less IE than Beryllium because there is a shift from s-orbitals to p-orbitals, which are more shielded, thus weaker nuclear attraction so lower IE

N → O

Oxygen's outermost e^- is paired in a p-orbital, and since paired e^- repel each other, less energy is needed to remove the electrons

• down a group

→ atomic radius ↑ [larger atomic radius]

→ shielding ↑ [more shells]

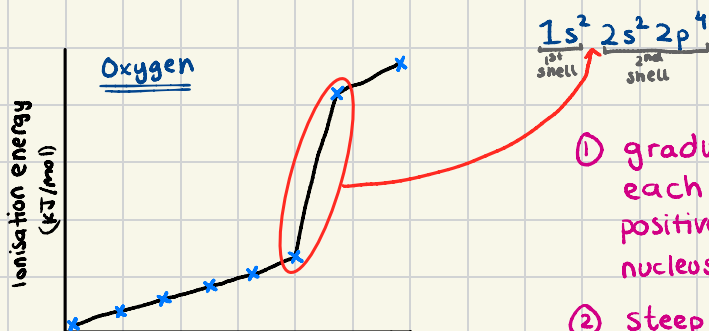
→ charge ↑ [more protons]

} lowers IE
} increases IE

atomic radius + shielding outweighs charge

TREND : IE decreases down the group

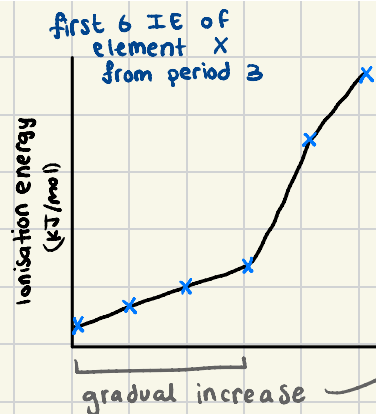
Successive ionization energy



∴ shows the energy required to remove each electron

① gradual increase in IE between each e^- is because each time an e^- is removed, the ion becomes more positive, thus there is a greater force of attraction between nucleus & e^-

② steep increase in IE from 6^{th} to 7^{th} e^- is due to a new shell starting which is much closer to the nucleus thus greater force of attraction between nucleus & e^-



How do we deduce what element X is?

this means there are 4 electrons in the outermost shell

by observing the periodic table, we can deduce that element X, which is in period 3 & group 4, is silicone (Si)

Past paper Q

Elements X and Y are in the same group of the periodic table.

The table shows the first six ionization energies of X and Y in kJ/mol:

	1st	2nd	3rd	4th	5th	6th
X	800	1600	2400	4300	5400	10400
Y	1000	1800	2700	4800	6000	12300

steep increase

X & Y have 5 outershell electrons

↳ * Group 5 * ⇒ arsenic and antimony
but which is X & Y?

What could be the identities of X and Y?

	X	Y
A	antimony, Sb	arsenic, As
B	arsenic, As	antimony, Sb
C	selenium, Se	tellurium, Te
D	tellurium, Te	selenium, Se

↑
element Y required more energy than X

X has to have a bigger atomic radius (one more shell) which is why it required less energy. This means X is in period 5 & Y is in period 4.

↳ * X is antimony & Y is arsenic *